Report on

The target-free clique conjecture for threshold-linear networks: complete exact maps for $n \leq 7$, the $\varepsilon \to 0$ map at $n = 8$, infinite families, and unbounded thresholds

This report is based on the supplied 34-page revised draft dated October 2026. The repository materials, including code, certificates and deferred proof notes, were not reviewed.

Potentially important issues

None identified in this paper-based review.

Very minor points

1. The two-parameter boundary list omits off-support curves (Section 3.1, page 8). The paragraph beginning "The two-parameter version" says that the SFP verdict is invariant on connected components of the complement of $a_i = 0$, $D = 0$, $\Delta_{m-1} = 0$ and $L = 0$. For a proper support σ, this assertion fails: the outside-neuron condition can change sign without crossing any of the listed curves. Lemma 3.2 and the later definition of Z include the polynomials o_k for $k \notin \sigma$, but this opening list does not.

For a concrete example, let $\sigma = \{1, \dots, 6\}$ induce G_1 from Table 1, and add vertex 7 receiving edges from exactly $2, 3, 5, 6$, with no outgoing edges. Set $t = 100$ and let β range over $[52/25, 209/100]$. Throughout this segment, all six a_i, D, all six Hurwitz determinants and L are strictly positive. Exact root counts give no zero of any of these polynomials in the segment, and their endpoint signs are positive. The segment therefore lies in one connected component of the complement of the listed curves.

However, the outside-neuron numerator is

$$o_7(\beta) = d - \beta(a_1 + a_4) = -\beta^5 + 3\beta^4 - 3\beta^3 + 3\beta^2 - 2\beta + 1,$$

and its endpoint values are

$$o_7(52/25) = \frac{417793}{9765625} > 0, \qquad o_7(209/100) = -\frac{1006161749}{10^{10}} < 0.$$

Since $D > 0$, these are also the signs of the outside drive. Thus σ is not an SFP at the first endpoint and is an SFP at the second, contradicting the stated invariance.

This is a local omission in the written statement for proper supports. The reconstructed set later in the paragraph uses the complete Z. Full supports have no outside-neuron conditions, so this example does not establish an error in the full-support counts or threshold values, or in the implementation.